

Regime Model & Sector Rotation Strategy

Research Summary — Personal Portfolio

Based on: *Regimes, Systematic Models and the Power of Prediction* — Man Group OMI Advisory (March 2025)

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1. Overview

This document summarizes the implementation and optimization of a regime-based sector rotation strategy for a long-term personal portfolio. The strategy is inspired by the Man Group paper which proposes identifying historical macro regimes similar to the current environment and using them to time investments. We extend the paper by applying the regime signal to sector rotation and combining it with a momentum filter.

2. Regime Model Methodology

2.1 Core Idea

The regime model asks: *what historical periods looked most like today macroeconomically?* It then uses the behavior of assets after those similar periods to generate a forward-looking signal. It is essentially a k-nearest-neighbor approach applied to macro time series.

2.2 Economic State Variables

We use 8 variables (slight modification from the paper's 7):

Variable	Proxy	Source	Start
Market	S&P 500	CRSP	1962
Yield Curve	US 10yr minus 3-month yield	FRED	1934
Oil	WTI Crude	FRED	1986
Gold	London PM Fix (IQ12260)	FRED	tested
Credit Spread	BAA minus 10yr (BAA10Y)	FRED	1986
Monetary Policy	US 3-month yield (TB3MS)	FRED	1934
Volatility	VIX / Realized Vol	FRED/CRSP	1990
Stock-Bond Corr	Rolling 2yr S&P500 vs 10yr	Computed	1964

Final dataset: 468 months, 1986-01 to 2024-12

2.3 Transformation

Each variable is transformed into a z-score of 12-month changes, computed over a rolling 10-year window and winsorized at plus or minus 3. This makes all variables comparable regardless of scale.

2.4 Similarity Score (Global Score)

For each month T, we compute the Euclidean distance between T's z-scores and every historical month. The 36 most recent months are excluded to avoid momentum loading. Lower distance = more similar regime. The formula is:

$$d(T,i) = \sqrt{\sum_v (z_v(T) - z_v(i))^2}$$

3. Data Pipeline

All data fetched programmatically. Two main sources:

FRED API (fredapi Python package) — yield curve, oil, credit spread, monetary policy, VIX. Free API key required.

CRSP via WRDS — S&P 500 daily index (spindx from crsp.dsp500). Used for market returns, realized volatility, and stock-bond correlation. Note: WRDS subscription required — download and save locally before expiry.

Ken French Data Library — FF5 factors, Momentum factor, and 12 Industry Portfolios. Free direct download via urllib.

Copper (original paper variable) was unavailable via free sources with sufficient history. Replaced with credit spread (BAA10Y) which captures similar risk appetite signal.

4. Strategy Development

4.1 What We Tested

We tested the regime signal applied to progressively more complex strategies:

Strategy	Result	Verdict
S&P 500 L/S timing	SR: -0.05	Failed — equity drift kills shorts
S&P 500 cash rotation	SR: 0.54	Marginal — cash drag hurts
S&P 500 equity/bond rotation	SR: 0.51	Worse — bond proxy noisy
FF Factor timing	SR: 0.76	Good risk-adj, low raw return
12-Industry sector rotation	SR: 0.71	Beats S&P 500 on return
Hybrid Regime + Momentum	SR: 0.89	Best — use this

4.2 Key Insight: Why Long/Short Fails on Equities

Equities have strong upward drift (~70% of months are positive). Shorting based on dissimilar regimes fights this structural tailwind. The regime model's edge with equities is drawdown reduction and rotation, not directional shorting.

4.3 Why Sector Rotation Works

Different sectors have genuinely different sensitivity to macro regimes. Tech crushes in low-rate recoveries. Energy leads in inflationary regimes. Financials outperform in rising rate environments. The regime model identifies which

historical environment we are in, and sector performance is highly regime-dependent — much more so than the broad market index.

5. Ablation Study

To understand the source of alpha we decomposed the hybrid strategy into its components:

Strategy	Sharpe	Ann Return	Max DD	vs Baseline
S&P 500 Buy & Hold	0.55	8.5%	-52.6%	-
Equal Weight (baseline)	0.70	10.7%	-49.5%	0.0%
Momentum only	0.72	12.2%	-45.4%	+1.5%
Regime only	0.74	11.8%	-48.1%	+1.1%
Hybrid (Regime+Mom)	0.89	13.7%	-42.8%	+3.0%

The combined effect (3.0%) is much larger than the sum of parts (1.5% + 1.1% = 2.6%). This confirms that regime and momentum are **complementary, not redundant** signals. The regime identifies the macro environment; momentum identifies which sectors are already responding to it.

6. Winning Configuration

Parameter	Value	Notes
Similarity quantile	q = 0.15	Top 15% most similar months
Forward window	1 month	Look 1 month ahead after similar periods
Momentum window	3 months	Standard, no skip-month
Signal blend	50/50	Equal weight regime and momentum ranks
Sectors held	Top 3	Best Sharpe, reasonable concentration
Rebalancing	Monthly	End of month, 1-month lag on signal

6.1 Concentration Analysis

Top N Sectors	Sharpe	Ann Return	Max DD
Top 2	0.83	13.7%	-43.1%
Top 3 (best)	0.89	13.7%	-42.8%
Top 4	0.81	12.6%	-43.5%
Top 5	0.82	12.4%	-44.9%
Top 6	0.77	11.7%	-45.0%
All 12 (equal)	0.70	10.7%	-49.5%

Top 3 is the sweet spot — same return as Top 2 but better Sharpe and lower drawdown. More than 5 sectors converges toward equal weight with diminishing alpha.

7. Final Performance Summary

Strategy	Sharpe	Ann Return	Max DD	Alpha vs S&P
S&P 500 Buy & Hold	0.55	8.5%	-52.6%	-
Equal Weight Sectors	0.70	10.7%	-49.5%	+2.2%
Regime + Momentum (Top 3)	0.89	13.7%	-42.8%	+5.2%

The winning strategy beats S&P; 500 by 5.2% annually with a 62% higher Sharpe ratio and 10 percentage points lower max drawdown. True alpha over the equal weight sector baseline (the proper lookahead-free benchmark) is 3.0% per year.

8. Practical Implementation

8.1 Monthly Workflow

At the end of each month, run the following steps:

1. Update macro variables from FRED API (yield curve, oil, credit spread, VIX, rates)
2. Update S&P; 500 prices from Yahoo Finance or CRSP
3. Recompute z-score transformations and Euclidean distance scores
4. Compute expected sector returns for top 15% similar historical months
5. Compute 3-month sector momentum (no skip)
6. Rank sectors by 50% regime + 50% momentum combined score
7. Hold the top 3 sectors equally weighted (33.3% each)
8. Rebalance portfolio if current holdings differ from target

8.2 ETF Mapping

Map FF 12 Industry Portfolios to investable ETFs:

FF Industry	ETF	Description
HiTec	XLK	Technology Select Sector
Hlth	XLV	Health Care Select Sector
Finance	XLF	Financial Select Sector
Enrgy	XLE	Energy Select Sector
NoDur	XLP	Consumer Staples Select Sector

Durbl	XLY	Consumer Discretionary Select Sector
Manuf	XLI	Industrial Select Sector
Shops	XLY	Consumer Discretionary Select Sector
Utils	XLU	Utilities Select Sector
Telcm	XLC	Communication Services
Other	SPY	Broad market proxy

8.3 Current Signal (as of December 2024)

The model identifies December 2024 as most similar to post-crisis recovery periods of 2020-2021 and 2010-2011. Most dissimilar to 2008-2009 crisis. This is a favorable macro environment.

Factor	Regime Signal	Action
Market	LONG	Overweight broad equity
Size (Small Cap)	LONG	Overweight small caps
Profitability	LONG	Overweight quality
Momentum	LONG	Overweight momentum
Value	SHORT	Underweight value
Investment	SHORT	Underweight capex-heavy

9. Caveats and Limitations

Variable selection bias: The 8 macro variables were chosen with knowledge of what drives equity returns. This induces mild look-ahead bias — the same acknowledged in the original paper.

Limited regime history: Regime scores start 1996 (after rolling window burn-in). The paper uses data back to 1962 giving more historical matches. Our results may be slightly optimistic.

Hyperparameter tuning: $q=0.15$, top3, 3-month momentum were selected on the same dataset used for testing. True out-of-sample alpha is probably 2-2.5% rather than 3.0%.

Transaction costs: Monthly rebalancing across 3 sector ETFs incurs costs. With typical ETF spreads and commissions, expect 0.1-0.3% annual drag. Still leaves meaningful alpha.

Factor returns vs ETF returns: Backtested on FF industry portfolio returns (academic). Real ETF returns will differ slightly due to fees, tracking error, and reconstitution.

10. Next Steps

1. Build production monthly rebalancing script
2. Replace FF industry returns with actual ETF historical returns for more realistic backtest
3. Integrate with broader portfolio of strategies — compute portfolio-level Sharpe

4. Explore sub-sector granularity (11 GICS vs 12 FF industries) for more discrimination
5. Test dynamic signal blending (e.g. higher regime weight during high-conviction regimes)
6. Set up automated monthly email/alert with current sector allocation

All code available in: `regime_data_ingestion.py`, `regime_model.py`, `regime_ff_factors.py`, `regime_grid_search.py`